

Team 7: AVION, A3

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I. Drag Build-up

A. Drag Calculations

Quantity	Symbol	Value	Source / Notes
Aspect ratio	AR	3.5	OpenVSP (main wing geometry)
Parasite drag (clean)	$C_{D0, \text{clean}}$	0.0190	OpenVSP Parasite Drag tool
Oswald efficiency (clean)	e_{clean}	0.825	[1] (range 0.80–0.85)
Oswald efficiency (takeoff flaps)	e_{TO}	0.775	[1] (range 0.75–0.80)
Oswald efficiency (landing flaps)	e_{L}	0.725	[1] (range 0.70–0.75)
Induced drag factor (clean)	K_{clean}	$\frac{1}{\pi e_{\text{clean}} AR}$	
Induced drag factor (takeoff)	K_{TO}	$\frac{1}{\pi e_{\text{TO}} AR}$	
Induced drag factor (landing)	K_{L}	$\frac{1}{\pi e_{\text{L}} AR}$	

Table 1: First-order drag inputs and derived induced-drag factors.

B. Drag Polars

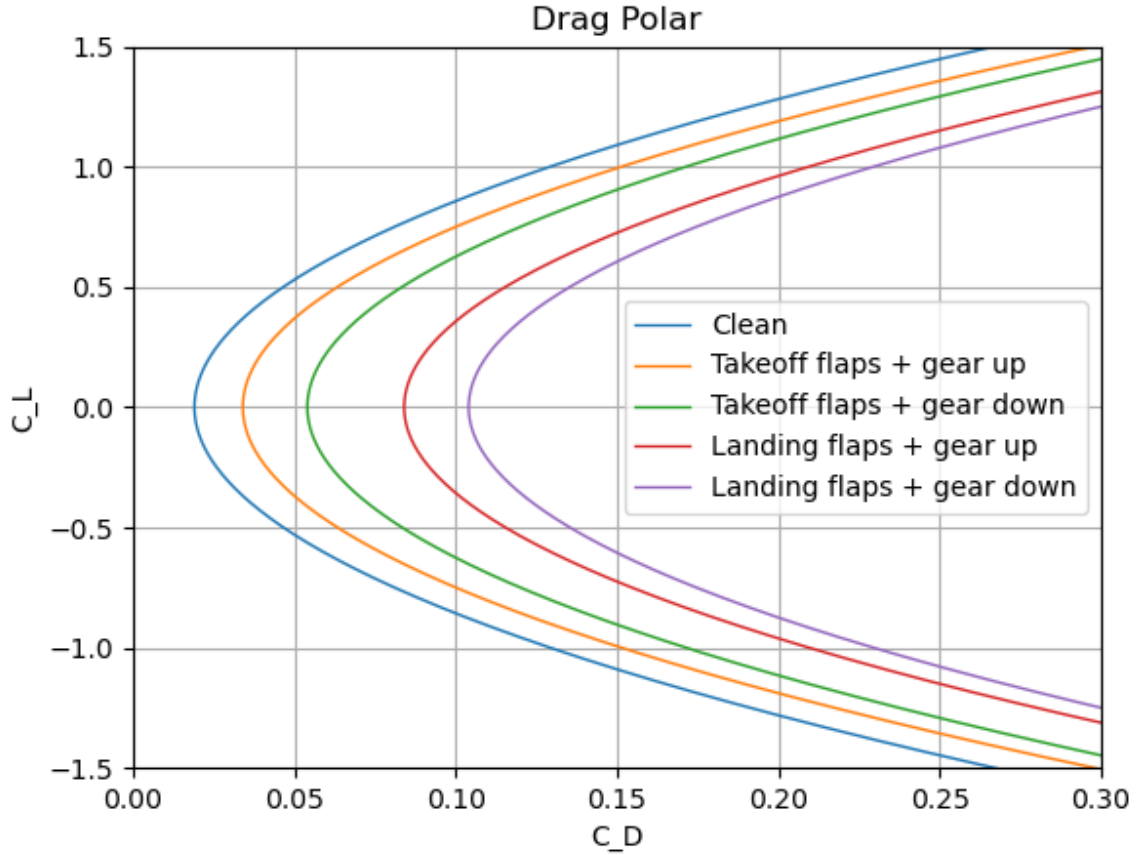


Figure 1: Drag Polar Plot

C. Assumptions

A first order drag build up approach was used for this analysis. The drag polar was modeled as parabolic, with total drag being the sum of induced and parasite drag. The C_{D_0} was taken from OpenVSP's Parasite Drag tool and treated as constant with lift coefficient. Induced drag was approximated using a finite wing approximation with a constant Oswald efficiency factor and aspect ratio. The Oswald efficiency values for takeoff, landing, and clean configurations were chosen from typical ranges [2] to account for higher flow separation and non-elliptic lift distributions created by flap deflection. Gear down configurations were modeled by using higher parasite drag and maintaining the same induced drag design. Reynolds number variation, viscous inviscid interaction effects, and compressibility effects were assumed negligible due to this analysis' main objective being preliminary sizing as opposed to high fidelity aerodynamic assessment.

II. Constraint Diagram

A. Identification of Constraint Equations

The aircraft sizing process is governed by performance constraints derived from the requirements and design objectives. Each constraint limits the allowable combinations of thrust-to-weight ratio (T/W) and wing loading (W/S). The relevant constraints considered in this analysis are summarized below and are plotted directly on the T/W – W/S diagram. All values were taken from their given units (see Table 2) and converted in the ft-slug-s unit system for analysis. All equations were taken from either Harvey or Raymer [1][3].

1. Stall Constraints

This constraint ensures the safe operation during low speeds, such as during takeoff and landing. Each regime has different stall constraints; the section is split into landing, cruise, and takeoff. Each regime has different lift coefficients and thus different stall speeds.

$$\frac{W}{S} = \frac{1}{2} \rho V_{stall}^2 C_{L_{max}} \quad (1)$$

The three different speeds and maximum lift coefficients for stall constraints are listed on Table 2.

2. Takeoff Field Length Constraint

Ensures compliance with the maximum allowable takeoff distance.

$$\left(\frac{W}{S}\right)_{TO} = \frac{0.5 \rho (V_{WOD} + V_{cat})^2 \cdot C_{L_{max_{TO}}}}{1.21} \quad (2)$$

where V_{WOD} is the "wind over deck airspeed", or the relative airspeed of the carrier itself. V_{cat} is catapult launch airspeed.

3. Landing Field Length Constraint

Limits maximum wing loading to ensure acceptable landing distance.

$$\left(\frac{W}{S}\right)_L = \frac{1}{2}\rho V_{app}^2 C_{L_{\max L}} \quad (3)$$

This constraint appears as a vertical line on the T/W versus W/S plot.

4. Climb Gradient Constraint

Enables the aircraft to be capable of climbing at a rate acceptable to the design requirements.

$$\left(\frac{T}{W}\right)_{climb} = \frac{D}{W} + G \quad (4)$$

Define G as the climb gradient ($ft_{\text{alt gain}}/ft_{\text{forw. travel}}$) and let drag be evaluated as a function of wing loading:

$$\frac{D}{W} = \frac{qC_{D0}}{W/S} + \frac{K}{q} \left(\frac{W}{S}\right) \quad (5)$$

Recall

$$q = \frac{1}{2}\rho V^2 \quad (6)$$

is the dynamic pressure. Define a constant K as the induced drag factor, used in the drag polar and for drag calculations in initial sizing.

$$K = \frac{1}{\pi e(AR)} \quad (7)$$

Recall e is the Oswald efficiency factor, which is function of the lift distribution along the wingspan. The aspect ratio approximation was calculated from the initial sketch in OpenVSP.

5. Ceiling Constraint

Ensures a safe minimum T/W value in level flight, which corresponds to a maximum operational altitude of the aircraft.

$$\left(\frac{T}{W}\right)_{min} = 2\sqrt{KC_{D0}} \quad (8)$$

6. Cruise Constraint

Ensures sufficient thrust is available to overcome drag at cruise speeds specified by the design requirements.

$$\left(\frac{T}{W}\right)_{cr} = \frac{qC_{D0}}{W/S} + \frac{K}{q} \left(\frac{W}{S}\right) \quad (9)$$

7. Maneuverability (Sustained Turn) Constraint

The maneuverability constraint ensures the aircraft can sustain a specified load factor n at a given speed and altitude without stall. As stall in a maneuver comes at a higher speed than level flight, this is a more stringent constraint than the stall constraints in level flight for takeoff, cruise, and landing presented above. This requirement is derived from the sustained turn rate requirement provided in the U.S. Navy’s RFP [4].

$$\left(\frac{T}{W}\right)_{man} = \frac{qC_{D0}}{W/S} + \frac{K}{q}n^2 \left(\frac{W}{S}\right) \quad (10)$$

Note that n is the sustained load factor, in multiples of g ’s.

8. Arrestment/ recovery constraint

The arrestment speed requirement was estimated from historical data of stall speed using the standard carrier approach relationship. A stall speed of $V_S = 195$ ft/s (115 KTAS) is used from the Boeing F/A-18C/D Hornet [5]. Applying the relation $V_{arr} \approx 1.15 V_{stall}$ [1], the resulting arresting speed is approximated as 132 KTAS. To check compatibility with the arresting gear, the landing weight was estimated using fuel fractions obtained in A1. A fuel fraction of $W_{fuel}/W_0 = 0.387$ corresponds to a landing weight with full payload capacity loaded and 20% fuel remaining, a conservative estimate. Using this, the aircraft landing weight becomes $(1 - 0.387)W_0$, which for current TOGW corresponds to 31,000 lbs. Comparing this value with the arresting envelope provided for the Gerald R. Ford system shows that our design falls within acceptable region and meets with 100,000 lb hook-load, enabling safe landing of the aircraft.

B. Input parameters

Parameter	Symbol	Value	Units	Source
Cruise Mach number	M	0.85	–	[4]
Cruise altitude density	ρ	8.91	lbs/ft ³	[6]
Zero-lift drag coefficient	C_{D0}	0.0190	–	–
Induced drag factor	K	0.114	–	–
Max lift coefficient (TO)	$C_{L_{max_{TO}}}$	2.0	–	[3]
Max lift coefficient (L)	$C_{L_{max_L}}$	2.3	–	[3]
Max lift coefficient (cr)	$C_{L_{max_{CR}}}$	1.5	–	[3]
Required climb gradient	G	0.22	–	–
Sustained load factor	n	7–8	g	[2]
”Wind Over Deck” airspeed	V_{WOD}	51	ft/s	[4]
Airspeed from Catapult	V_{CAT}	200	ft/s	[4]

Table 2: Input parameters and assumptions used for constraint analysis

Assumptions:

A parabolic drag polar is used, compressibility effects are neglected in preliminary sizing, and thrust lapse with altitude is ignored. Takeoff and landing lift coefficients are assumed achievable with high-lift devices deployed. A thrust cruise to takeoff ratio was calculated based on F414 GE engine specifications, where wet thrust of 22,000 lbf was used for cruise and dry thrust of 13,000 lbf was used for takeoff. All sources in Table 2 that are listed as – were self-calculated.

C. Graph Output

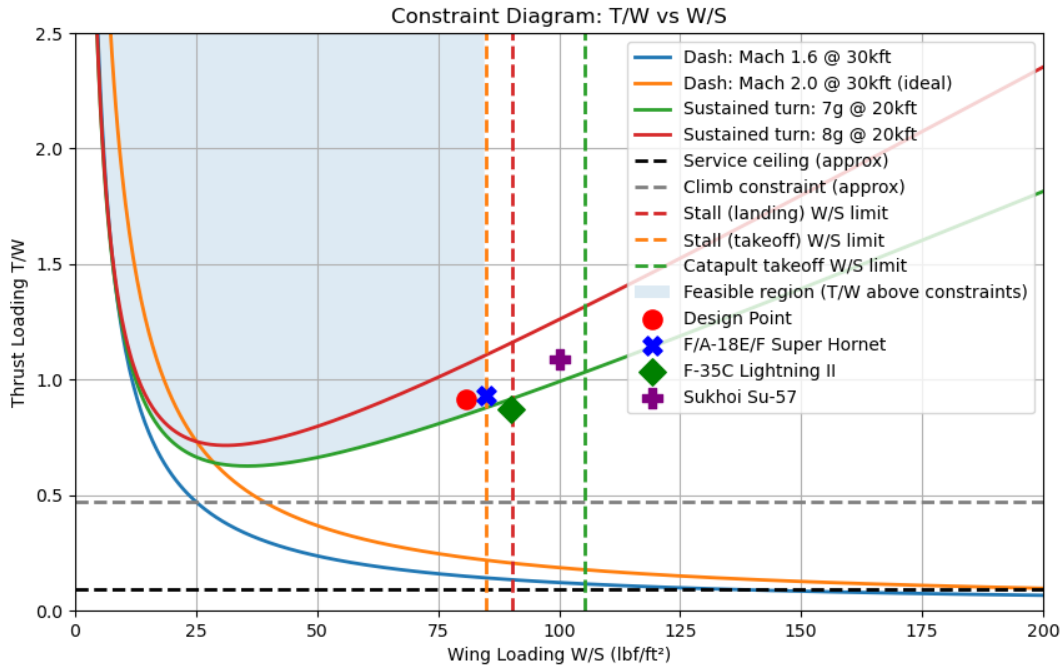


Figure 2: Constraint Diagram with design points and comparable aircraft

D. Design Point Selection

A design point of $W/S = 80.7$ and $T/W = 0.91$ is selected, as it satisfies all performance constraints while minimizing thrust requirements and maintaining a reasonable wing area. This point provides adequate margin for climb performance and takeoff distance while remaining compatible with cruise efficiency objectives outlined in the RFP [4].

References

- [1] D. P. Raymer, *Aircraft Design: A Conceptual Approach*. AIAA Education Series, American Institute of Aeronautics and Astronautics, 2 ed., 1992.

- [2] J. Roskam, *Airplane Design*. University of Kansas, 1 ed., 1999.
- [3] C. Harvey, *A Metabook for Aircraft Design*. Department of Mechanical and Aerospace Engineering, University of California, Davis, 1 ed., 2026.
- [4] D. Wells, “Request for proposal: Next generation carrier-based strike fighter aircraft,” tech. rep., AIAA Aircraft Design Technical Committee, 2026.
- [5] *NATOPS FLIGHT MANUAL NAVY MODEL F/A-18A/B/C/D*, 2008.
- [6] *US STANDARD ATMOSPHERE*, 1976.

III. Appendix

A. CRediT Statement

Charlie Stone: Developed code for Constraint Diagram, formatted and edited report

Mostafa Hashem: Developed code for Drag Polar and polished C.D. code

Valeria Cecena: Contributed to report writing

Quinn Kennerly: Primary drafting of report

Lena Pattamadilok: Contributed to report writing

Jose Hernandez Negrete: Contributed to report writing

B. AI and Code Use Statement

Python programming language was used. AI was used to debug code. ChatGPT was implemented to check grammar and for debugging of BibLaTeX .bib file formatting.

C. Source Code

Both the drag buildup code and constraint diagram code are provided in text form in this report, as a file submission, and at GitHub at the following link under

EAE130SP_AVION7/Assignments/A3:

https://github.com/charliesstone/EAE130SP_AVION7.git

```
import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Inputs you must set

AR = 4.5 # wing aspect ratio OpenVSP

CD0_clean = 0.0190 # Clean CD0 OpenVSP
```

```

#delta CD0 from slides
dCD0_takeoff_flaps = 0.015    # -0.0100.020
dCD0_landing_flaps = 0.065    # -0.0550.075
dCD0_gear           = 0.020    # -0.0150.025

# Oswald e by stage from slides
e_clean    = 0.825    # -0.800.85
e_takeoff  = 0.775    # -0.750.80
e_landing  = 0.725    # -0.700.75

# CL max assumptions from raskom
CLmax_clean    = 1.6
CLmax_takeoff  = 2.0
CLmax_landing  = 2.3

# configuration table

configs = np.array([
    "Clean",
    "Takeoff flaps",
    "Takeoff flaps",
    "Landing flaps",
    "Landing flaps"
])

dCD0_flaps = np.array([
    0.0,
    dCD0_takeoff_flaps,
    dCD0_takeoff_flaps,
    dCD0_landing_flaps,
    dCD0_landing_flaps
])

dCD0_gear = np.array([
    0.0,
    0.0,
    dCD0_gear,
    0.0,
    dCD0_gear
])

e_values = np.array([
    e_clean,
    e_takeoff,
    e_takeoff,
    e_landing,
    e_landing
])

CL_max = np.array([
    CLmax_clean,
    CLmax_takeoff,
    CLmax_takeoff,

```

```

        CLmax_landing,
        CLmax_landing
    ])

# pandas dataframe
config_df = pd.DataFrame({
    "config": configs,
    "dCDO_flaps": dCDO_flaps,
    "dCDO_gear": dCDO_gear,
    "e": e_values,
    "CL_max": CL_max
})

# Compute CDO for each configuration using delta CDO
config_df["CDO_config"] = CDO_clean + config_df["dCDO_flaps"] + config_df["
    dCDO_gear"]

# induced-drag factor k for each configuration (since e changes with flaps)
config_df["k"] = 1.0 / (math.pi * AR * config_df["e"])

print("\n== Final CDO per configuration (Clean + ΔCDO) ==")
print(config_df[["config", "CDO_config", "dCDO_flaps", "dCDO_gear", "e", "k"]]
    .to_string(index=False))

# drag polar plot

plt.figure()

for _, row in config_df.iterrows():
    CL = np.linspace(-3.0, row["CL_max"], 900) # CL from -4 to CLmax
    CD = row["CDO_config"] + row["k"] * (CL**2)
    plt.plot(CD, CL, linewidth=1, label=row["config"])

plt.xlabel("C_D")
plt.ylabel("C_L")
plt.xlim(0, 0.30)
plt.xticks([0, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30])
plt.ylim(-1.5, 1.5)
plt.title("Drag Polar")
plt.grid(True)
plt.legend()
plt.show()

```

Listing 1: Drag Polar, Python

```

import numpy as np
import matplotlib.pyplot as plt

## ----- EAE 130A, preliminary design sizing ----- ##
## ----- Team 7 Avion: Mostafa Hashem, Charlie Stone, Jose Hernandez Negrete,
## ----- ##
## ----- Valeria Cecena, Quinn Kennerly, Lena Pattamadilok ----- ##

```

```

#region inputs
g = 32.174 #ft/s^2, gravitational acceleration
tar_togw = 50,000 #lbf, target gross take off weight
S_i = 600 #ft^2, initial wing area assumption including strakes (LERX)
S_ialt = 400 #ft^2, initial wing area assumption not including strakes (LERX)
C_D0 = 0.01755 #initial zero-lift drag estimation, from OpenVSP, see:
#(https://github.com/charliesstone/EAE130SP_AVION7/blob/main/Assignments/A2/
    Concepts/concept_1_COYOTE.vsp3)
e = 0.8 #oswald efficiency factor for fighters, from raymond textbook
AR = 3.5 #aspect ratio, from OpenVSP main wing, not including strakes (LERX)
nz_tar = 7.0 * g #ft/s^2, load factor, minimum targeted
nz_idl = 8.0 * g #ft/s^2, load factor, ideal targeted
wingload = np.linspace(0, 200, 500)
thrustload = np.linspace(0, 2.5, 500)
VWOD = 51 #ft/s (30 kts) "Wind Over Deck" airspeed, or wind speed + carrier
    cruise velocity
VCAT = 200 #ft/s (120 kts) airspeed generated by forward throw of CATapult
k = 1/(np.pi * e * AR) #drag polar constant, see 07-PreliminarySizing_Part3.
    pdf pg15 in canvas files
#endregion

#region stall
Vstall_L = 195 #ft/s (115 KIAS), stall speed for cruise (based on F/A-18E/F)
Vstall_C = 220 #ft/s (130 KIAS), stall speed for landing (based on F/A-18E/F)
Vstall_T = 205 #ft/s (120 KIAS), stall speed for takeoff (based on F/A-18E/F)
rho_30k = 8.91E-4 #slugs/ft^3, atmospheric density at 30k ft
rho_SL = 23.77E-4 #slugs/ft^3, atmospheric density at sea level
CLmax_L = 2.0
CLmax_C = 1.5
CLmax_T = 1.7

def stall(V, CL, rho):
    WS = (rho * CL * V**2)/2
    return WS
stallWS_L = stall(Vstall_L, CLmax_L, rho_SL) #wing loading constraint for
    stall on landing
stallWS_C = stall(Vstall_C, CLmax_C, rho_30k) #wing loading constraint for
    stall on cruise
stallWS_T = stall(Vstall_T, CLmax_T, rho_SL) #wing loading constraint for
    stall on takeoff
#endregion

#region takeoff
def takeoff(rho, Vwod, Vcat, CLMT):
    WS = (0.5 * rho * (Vwod + Vcat)**2 * CLMT)/1.21
    return WS
takeoffWS = takeoff(rho_SL, VWOD, VCAT, CLmax_T) #wing loading constraint for
    clearing carrier runway on takeoff

#endregion

#region climb
#assume CLmax_climb = CLmax_T (takeoff)

```

```

def climb(k, Cd0, CLMCLB):
    ks = 3.4 #Vclimb ~ Vstall proportionality constant (using Vclimb 748 ft/s,
    MROC for Super Hornet, and Vstall_C)
    G = 0.220 #ft/ft (unitless) Climb Gradient, = sin(g) ~= g, where g = climb
    pitch angle
    tempoverinc = 1/(0.8) #loss of thrust due to 50F increase over standard
    temperature
    maxcont2max = 1/(0.94) #ratio of maximum continuous to maximum peak thrust
    TW_gen = (ks**2 * Cd0)/(CLMCLB) + k * (CLMCLB)/(ks**2) + G
    TW = tempoverinc * maxcont2max * TW_gen #other parameters not included,
    see 07-PreliminarySizing_Part3.pdf pg20
    return TW
climbTW = climb(k, C_D0, CLmax_T)
#endregion

#region ceiling
ceilingTW = 2 * np.sqrt(k * C_D0)
#endregion

#region climb/dash
M_cruiseidl = 2.0
M_cruise_tar = 1.6
def dash(Cd0, k, WS, M, rho):
    Wf_Wi_cruise = 0.7706 #Cruise/Takeoff weight fraction, from A2 sizing code
    Tcr_Tto = 22/13 #Cruise/Takeoff thrust fraction, from the GE F414 engine
    deck (note: refine value for A3)
    Vcruise = M * 996 #996 ft/s is speed of sound at 30kft from NASA standard
    atmosphere tables
    qcr = 1/2 * rho * Vcruise**2 #lbf/ft^2, dynamic pressure at cruise
    velocity
    WS_cruise = WS * Wf_Wi_cruise #wing loading at cruise, as opposed to
    takeoff
    TW = Wf_Wi_cruise/Tcr_Tto * ( (qcr * Cd0)/(WS_cruise) + k/qcr * (WS_cruise
    ) ) #thrust loading at takeoff as a function of wing loading at cruise ->
    wing loading at takeoff
    return TW
TW_cruiseMa2 = dash(C_D0, k, wingload, M_cruiseidl, rho_30k)
TW_cruiseMa1p6 = dash(C_D0, k, wingload, M_cruise_tar, rho_30k)

#region maneuverability
# assume sustained turn at 20kft
rho_20k = 0.001267 # slugs/ft^3
V_turn = 1000 # ft/s
q_turn = 0.5 * rho_20k * V_turn**2

def maneuver_TW(WS, Cd0, k, q, n):
    # Sustained turn : T/W = D/W at load factor n
    return (q * Cd0) / WS + (k * n**2 * WS) / q

#target and ideal load factors
n_tar = 7.0 # target Nz
n_idl = 8.0 # ideal Nz

TW_manuv7g = maneuver_TW(wingload, C_D0, k, q_turn, n_tar)

```

```

TW_manuv8g = maneuver_TW(wingload, C_D0, k, q_turn, n_id1)
#endregion

#region constraint diagram plot
plt.figure(figsize=(10,6))

# --- Curves (T/W vs W/S) ---
plt.plot(wingload, TW_cruiseMa1p6, linewidth=2, label="Dash: Mach 1.6 @ 30kft"
)
plt.plot(wingload, TW_cruiseMa2, linewidth=2, label="Dash: Mach 2.0 @ 30kft
(ideal)")
# Maneuverability curves (choose whichever you implemented)
plt.plot(wingload, TW_manuv7g, linewidth=2, label="Sustained turn: 7g @ 20kft"
) # if Option A
plt.plot(wingload, TW_manuv8g, linewidth=2, label="Sustained turn: 8g @ 20kft"
) # if Option A
# plt.plot(wingload, TW_man_rate, linewidth=2, label="Sustained turn: 8 deg/s
@ 20kft") # if Option B
# Ceiling (horizontal line)
plt.hlines(ceilingTW, xmin=wingload.min(), xmax=wingload.max(),
          colors="k", linestyle="--", linewidth=2, label="Service ceiling (
approx)")

# Climb (constant line in your current formulation)
plt.hlines(climbTW, xmin=wingload.min(), xmax=wingload.max(),
          colors="gray", linestyle="--", linewidth=2, label="Climb
constraint (approx)")

# --- Vertical W/S limits ---
plt.axvline(stallWS_L, color="tab:red", linestyle="--", linewidth=2, label="
Stall (landing) W/S limit")
plt.axvline(stallWS_T, color="tab:orange", linestyle="--", linewidth=2, label=
"Stall (takeoff) W/S limit")
plt.axvline(takeoffWS, color="tab:green", linestyle="--", linewidth=2, label="
Catapult takeoff W/S limit")

# shaded area

TW_envelope = np.maximum.reduce([
    TW_cruiseMa1p6,
    TW_manuv7g, # or TW_man_rate if using option B
    np.full_like(wingload, climbTW),
    np.full_like(wingload, ceilingTW),
])

# Max allowable W/S is the minimum of your W/S limits
WS_max = min(stallWS_L, stallWS_T, takeoffWS)

mask = wingload <= WS_max
plt.fill_between(wingload[mask], TW_envelope[mask], 2.5, alpha=0.15, label="
Feasible region (T/W above constraints)")
#choosing design point

```

```

WS_design = 0.95 * WS_max # near right edge of feasible region
TW_required_at_WS = np.interp(WS_design, wingload, TW_envelope)

# Adding small margin so it's inside the feasible region
margin = 1.08
TW_design = margin * TW_required_at_WS

print("\n== Selected Design Point (bottom-right feasible) ==")
print(f"WS_design = {WS_design:.2f} lbf/ft^2")
print(f"TW_design = {TW_design:.3f}")
print("landing stall speed", Vstall_L, "ft/s")
# Plot the design point on the constraint diagram
plt.scatter(WS_design, TW_design, s=120, marker="o", color="red", zorder=10,
            label="Design Point")

#plot design point comparison with F/A-18E/F Super Hornet
WS_FA18 = 85.0 #lbf/ft^2, wing loading of
TW_FA18 = 0.93 #thrust loading of F/A-18E/F Super Hornet, from A2 sizing code
plt.scatter(WS_FA18, TW_FA18, s=120, marker="X", color="blue", zorder=10,
            label="F/A-18E/F Super Hornet")

#plot design point comparison with F-35C Lightning II
WS_F35 = 90.0 #lbf/ft^2, wing loading of F-35C Lightning II, from A2 sizing
code
TW_F35 = 0.87 #thrust loading of F-35C Lightning II, from A2 sizing code
plt.scatter(WS_F35, TW_F35, s=120, marker="D", color="green", zorder=10, label
            ="F-35C Lightning II")

#plot design point comparison with sukoi Su-57
WS_SU57 = 100.0 #lbf/ft^2, wing loading of Sukhoi Su-57, from A2 sizing code
TW_SU57 = 1.09 #thrust loading of Sukhoi Su-57, from A2 sizing code
plt.scatter(WS_SU57, TW_SU57, s=120, marker="P", color="purple", zorder=10,
            label="Sukhoi Su-57")
# plot

plt.xlim(0, 200)
plt.ylim(0, 2.5)
plt.xlabel("Wing Loading W/S (lbf/ft^2)")
plt.ylabel("Thrust Loading T/W")
plt.title("Constraint Diagram: T/W vs W/S")
plt.grid(True)
plt.legend(loc="upper right")
plt.show()

print("\n== Key W/S limits ==")
print("Landing stall W/S limit:", stallWS_L)
print("Takeoff stall W/S limit:", stallWS_T)
print("Catapult takeoff W/S limit:", takeoffWS)
print("Chosen WS_max:", WS_max)
#endregion

```

Listing 2: Constraint Diagram, Python